TinyStar'Ex

Build and assembly guide for the pointing/guiding system

Version 1.1

A STAROS project

staros-projects.org

1 Purpose

This document is the build, assembly and adjustment guide for the pointing and guiding system of the TinyStar'Ex (TSE) spectrograph. Figure 1 shows the assembled system with its guiding camera, here a Player One Ceres 462M, a well-suited model. This system complements the spectrograph part of TSE to form a complete set-up for observing the spectra of astronomical sources.

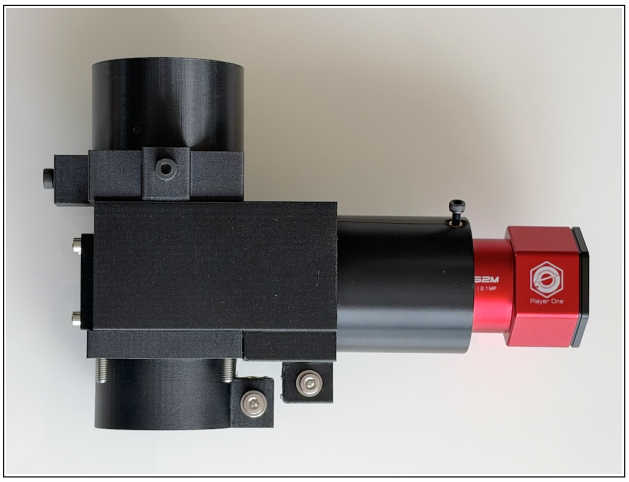


Figure 1. The TinyStar'Ex pointing and guiding system.

A video shows the main stages of the assembly (but do read this manual carefully too!):



Figure 2. Click on the image to watch the TinyStar'Ex assembly video: https://youtu.be/nrCZCETCJ2A

2 Licence

The TinyStar'Ex (TSE) project is released under an open licence that protects the rights of both users and authors. It keeps the project open and accessible, while respecting the work of all its contributors.

The TinyStar'Ex design files (STL print files, OpenSCAD sources and instruction manuals) are licensed under the *Creative Commons Attribution–NonCommercial–ShareAlike 4.0 International licence (CC BY-NC-SA 4.0)*. The optical kit, which is sold commercially, is not covered by this licence.

You are free to use, copy, modify and redistribute these files under the following conditions:

- Attribution: you must credit the authors (STAROS Projects) whenever you share or modify the files, indicate whether changes have been made, and provide a link to the licence;
- NonCommercial: the files may not be used for commercial purposes without prior permission from STAROS Projects. For example, selling parts printed from these files constitutes commercial use;
- ShareAlike: if you modify the files and distribute the result, you must do so under the same CC BY-NC-SA 4.0 licence.

The files are provided 'as is', without warranty of any kind, express or implied. The authors cannot be held liable for any damage arising from their use.

In short, anyone may freely download, print, adapt and share the TinyStar'Ex designs for personal, educational or non-profit use, provided the authors are credited and the designs are not exploited commercially.

A summary of the licence and its full legal text are available at: https://creativecommons.org/licenses/by-nc-sa/4.0/

3 The STL files

The STL files for printing the guiding-system parts are gathered in a compressed archive, which can be downloaded directly from the project's GitHub repository:

Download the guiding-system STL files:
stl_guide.zip

Figure 3 shows the parts to be printed to build the 'pointing/guiding' part of the TinyStar'Ex project.

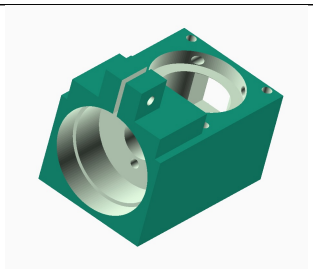
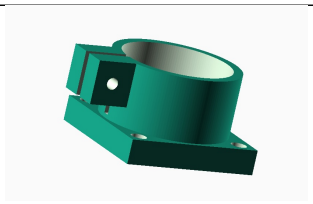


Figure 3. All the printed parts that make up the TinyStar'Ex pointing and guiding system.

Section 5 gives recommendations for printing these parts.

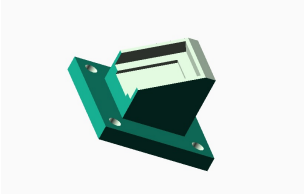
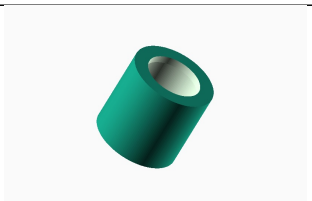
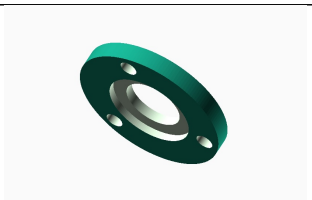
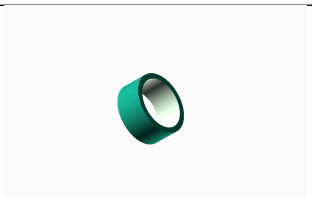
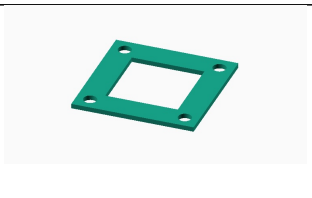
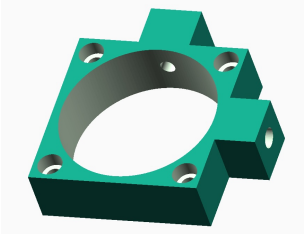
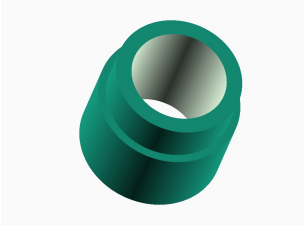
Table 1 is the bill of materials for the parts of the pointing/guiding system. The part names are those of the STL files.

Table 1. STL files of the TinyStar'Ex pointing/guiding system.

No.	Qty	View of the part
1G-1-cube-guidage	1	
2G-1-interface-guidage	1	

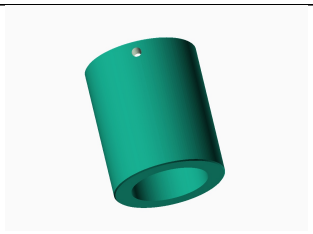
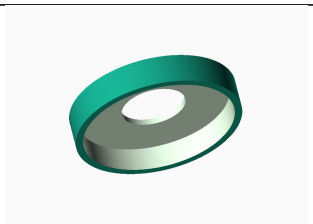
continued on next page...

Table 1 – continued from previous page

No.	Qty	View of the part
3G-1-support-miroir	1	
4G-1-tube-lentilles	1	
5G-1-bride-lentilles	1	
6G-1-entretoise-lentilles	1	
7G-1-entretoise-miroir	1	
8G-1-interface-telescope	1	
9G-1-coulant-50mm	1	

continued on next page...

Table 1 – continued from previous page

No.	Qty	View of the part
10G-1-interface-camera (stick camera version)	1	
11G-1-interface-camera (planetary camera version)	1	
12G-1-simulateur-f4	1	

Information on screws, fasteners and tools is given in the document describing the assembly of the spectrograph.

4 The source files

As with the spectrograph, you can modify the parts of the pointing and guiding system. Their OpenSCAD source files, `TSE_sources_guide.scad`, can be downloaded directly from the project's GitHub repository.

5 Printing

All the parts of the pointing and guiding system can be printed with a layer height of 0.18 mm.

Five perimeters are enough. You may go up to six for part `2G-1-interface-guidage`, which supports the whole spectrograph and therefore plays an important structural role. An infill of 30% with a gyroid pattern is entirely satisfactory. Supports for printing overhangs are needed only for parts `1G-1-cube-guidage` and `2G-1-interface-guidage` (see Figures 4 and 5).



Figure 4. Orientation of part `1G-1-cube-guidage` on the build plate, with its supports (here with the *Bambu Studio* slicer and a *Bambu Lab* printer). Note the use of 'tree' supports, needed to print this part correctly.

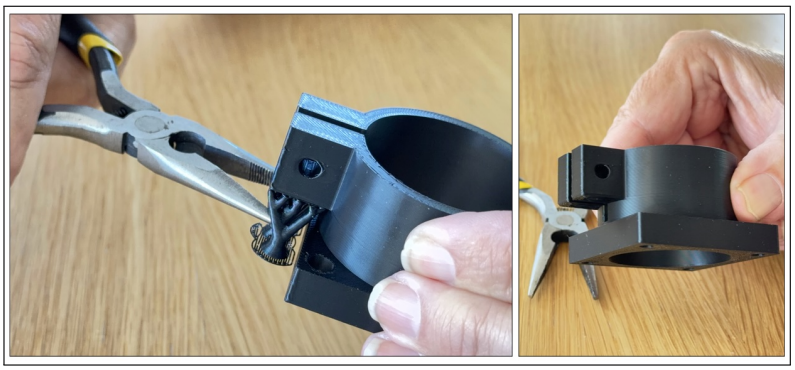


Figure 5. Removing the support used to print part `2G-1-interface-guidage`.

6 Assembling the pointing/guiding system

A number of inserts must be fitted in part `1G-1-cube-guidage`. Figures 6, 7 and 8 show their type and location.

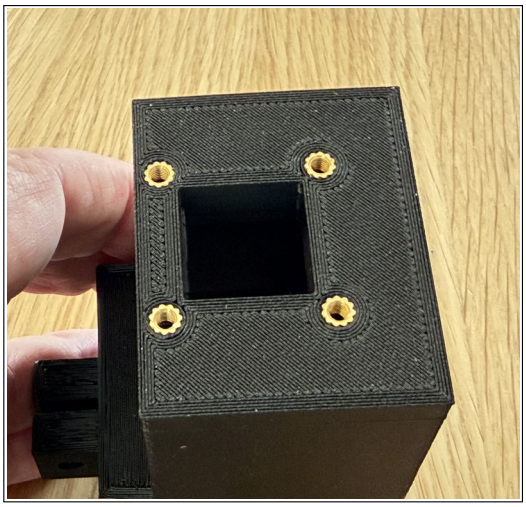


Figure 6. Inserts in part `1G-1-cube-guidage`: four M3 S (short) inserts.

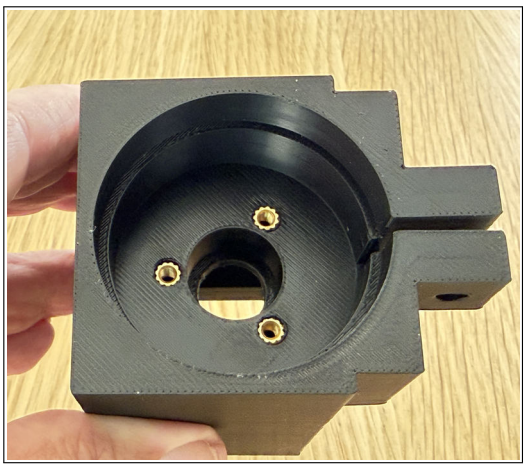


Figure 7. Inserts in part `1G-1-cube-guidage`: three M3 inserts.

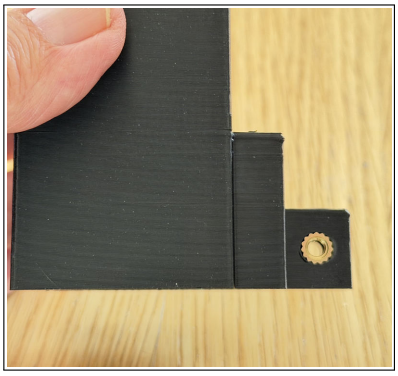


Figure 8. Insert in part `1G-1-cube-guidage`: one M4 S (short) insert.

While you are fitting inserts, you can also deal with those for the parts shown in Figures 9, 10 and 11.

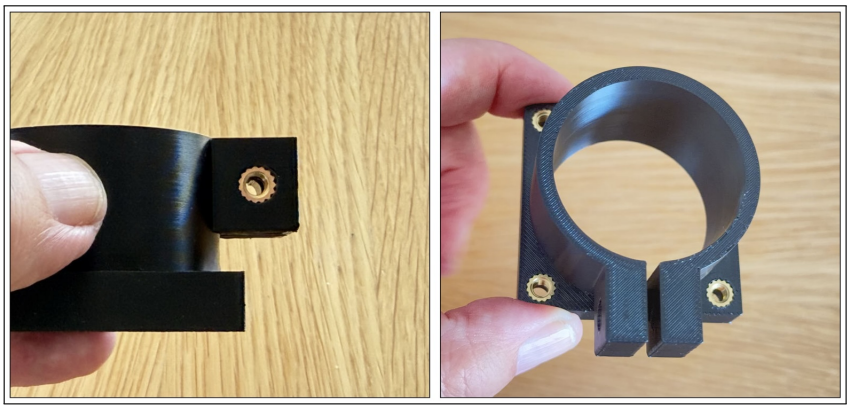


Figure 9. Inserts in part `2G-1-interface-guidage`: five M4 S (short) inserts.

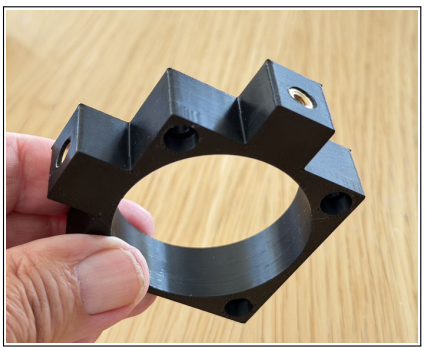


Figure 10. Inserts in part `8G-1-interface-telescope`: two M4 inserts.

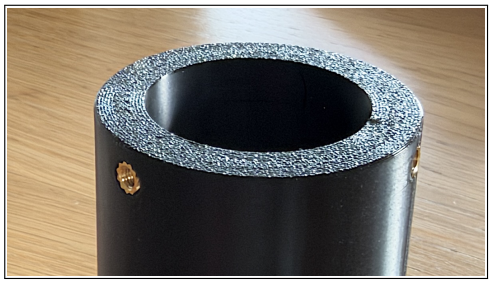


Figure 11. Inserts in part `10G-1-interface-camera`: two M3 inserts.

First of all, check that the guiding system interface fits properly over the collimator tube (Figure 12, left). The two parts must slide into each other without forcing. If they do not, the slit holder is probably not coaxial with the collimator tube (also look out for printing burrs, which are easy to remove by rubbing with abrasive cloth). Slightly loosen the three M2 screws holding the slit holder (Figure 12, right), then refit the guiding interface, pushing it fully home (yellow arrow in the left-hand view). With the interface still in place, retighten the three M2 screws of the slit holder. You can then remove the interface.

This operation is also an opportunity to see how some add-on parts are fixed: by clamping onto cylindrical bearing surfaces. This technique gives a firm hold without screws marking the printed parts. It also leaves great freedom for fitting accessories: the 40 mm diameter of the collimator tube (39.75 mm, to be precise) is a kind of 'standard' specific to TSE, onto which extensions are easily grafted.

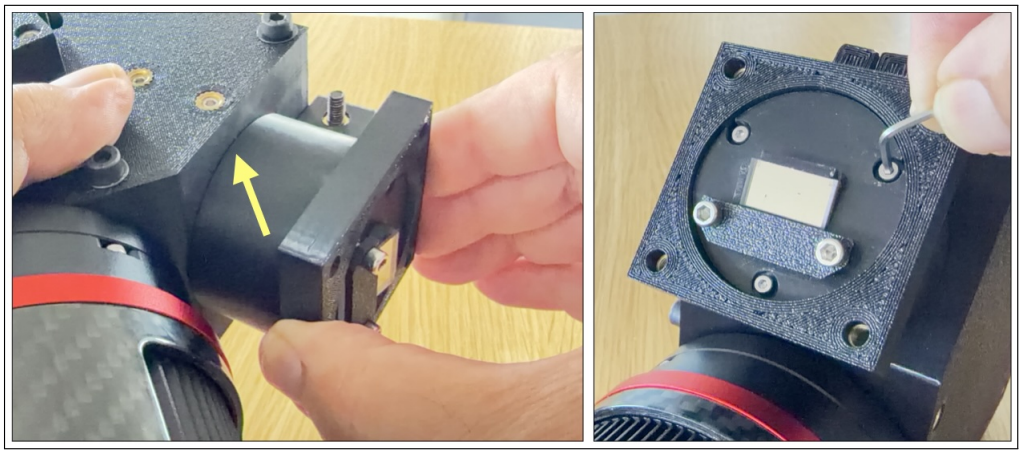


Figure 12. Checking that the collimator tube and the guiding system interface are coaxial. The two parts must meet at the yellow arrow.

Tip: the front face of the `2G-1-interface-guidage` part (Figure 13) marks precisely the longitudinal position of the slit, and therefore the plane in which the image formed by the telescope must be focused.

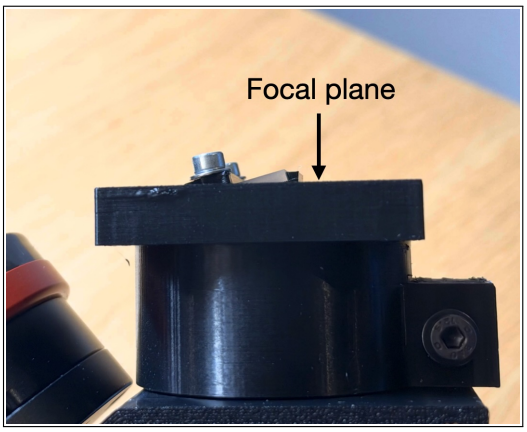


Figure 13. Location of the focal plane at the entrance of the spectrograph.

Next, fit the two doublets, 12.5 mm in diameter with a focal length of 50 mm, in the bore of part `4G-1-tube-lentilles` (Figure 14). They act as relay lenses, transferring the image of the slit to the detector plane of the pointing/guiding camera.



Figure 14. The components of the guiding optics sub-assembly: two lenses facing each other, the sleeve `4G-1-tube-lentilles`, part `6G-1-entretoise-lentilles` and the retaining part `5G-1-bride-lentilles`.

The lenses are fitted head to tail in their seat. It is essential to fit them the right way round: if you look carefully at them edge-on, you will see that one face is more curved than the other. The assembly must follow the orientations given in Figure 15 to obtain a good image in the guiding camera.

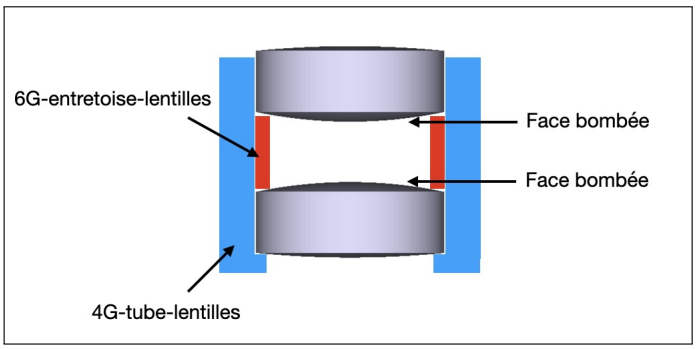


Figure 15. Assembly diagram of the guiding system lenses.

The lenses and the spacer should slide into their seat without difficulty. Insert the first lens (watch its orientation), then the spacer, then the second lens (again, watch the orientation). It is normal for the surface of the last lens to protrude from the tube, as shown in Figure 16.



Figure 16. The lenses in their seat. The outer face of the last lens must protrude slightly from the tube.

The tube fitted with the lenses is then mounted in its seat in the guiding cube (Figure 17).

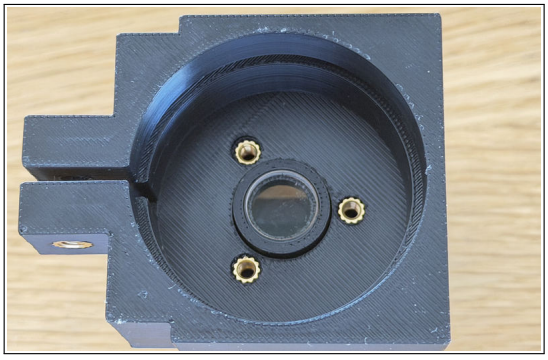


Figure 17. The lens tube in place in part `1G-1-cube-guidage`.

Next, fit part `5G-1-bride-lentilles`, held by three M3 screws 8 mm long (Figure 18). Tighten moderately, sharing the effort evenly between the three screws.

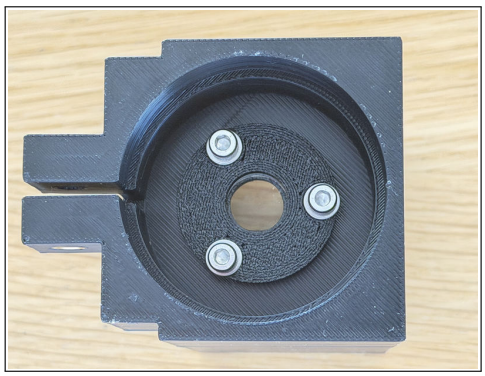


Figure 18. The lenses fixed in their seat. Take care to orient part `5G-1-bride-lentilles` so that its bore faces the lenses.

A 15 × 15 × 3 mm mirror sends to the guiding camera the image of the star field reflected by the outer parts of the slit, after the light has passed through the two lenses assembled previously.

This mirror is glued into part `3G-1-support-miroir`, a shoe inclined at 30° (see Figure 19).

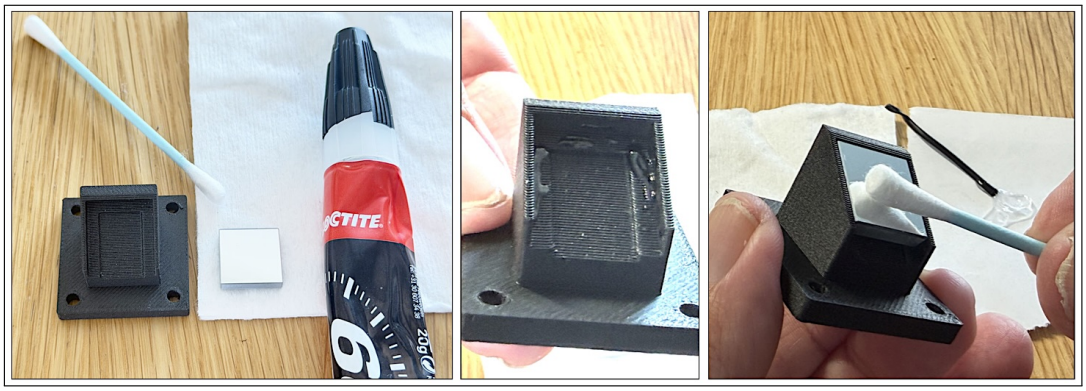


Figure 19. Assembling the fold mirror of the guiding system. An inexpensive, fairly fast-setting glue is used here. Apply very little glue to the areas on which the mirror rests (use a small stick or similar). Bring the mirror into position, pressed firmly against its stops, then press with a clean cotton bud for one minute. Leave the part to rest for a good hour before handling it.

Fix the fold mirror holder in the structure of the guiding block (Figure 20).

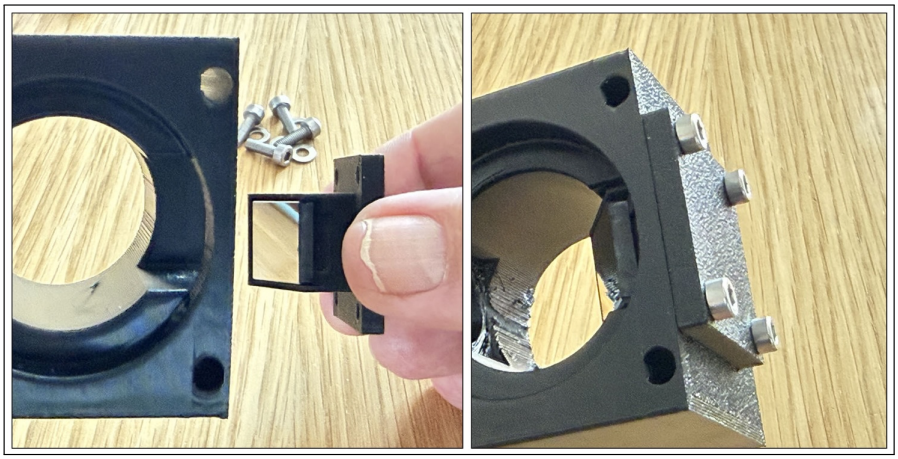


Figure 20. Fixing the flat fold mirror holder with four M3 × 10 mm screws. Do not fit it the wrong way round: refer to these photographs.

Next, fit the guiding camera interface. There are two options: a simple 3D-printed tube, part `10G-1-interface-camera`, or a more elaborate device, a helical focuser, here made by ZWO (Figure 21). We recommend the first option: it is the cheaper, and despite its apparent simplicity, it offers the best stability, both over time and whatever the region of sky at which the telescope is pointed.

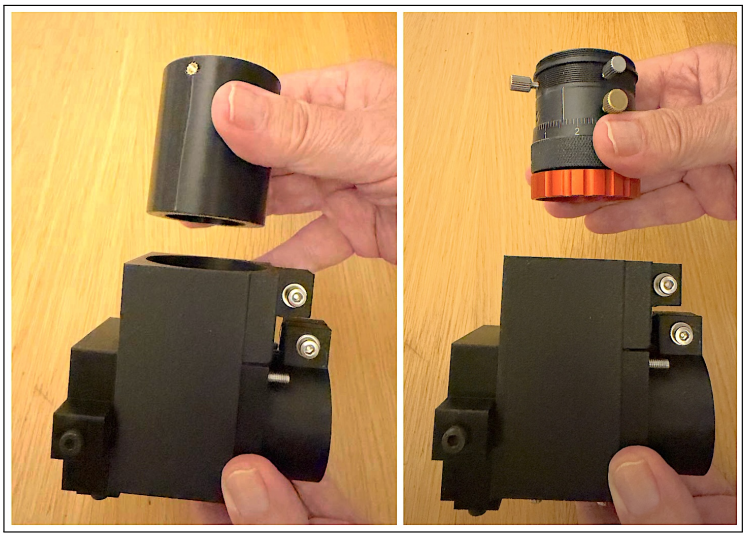


Figure 21. The two possible solutions for fitting the pointing and guiding camera.

Note also that the printed tube can take either a ‘stick’ camera or a ‘planetary’ camera (Figure 22), which the bulk of a helical focuser does not allow.

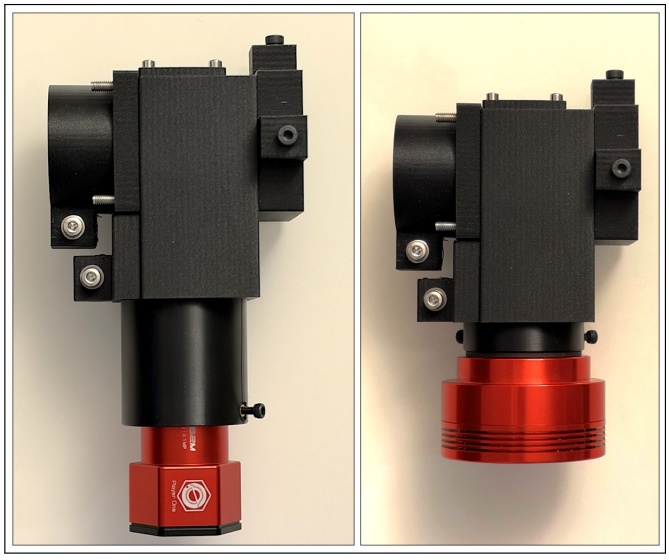


Figure 22. The pointing and guiding system is designed to accept both main families of camera. Only the length of the interface tube changes: the two configurations shown here differ by 38 mm, hence the two versions of the part, `10G-1-interface-camera` and `11G-1-interface-camera`.

In all cases, the interface is held by clamping: an M4 × 20 mm screw closes the slot of the collar to give a secure hold (Figure 23).

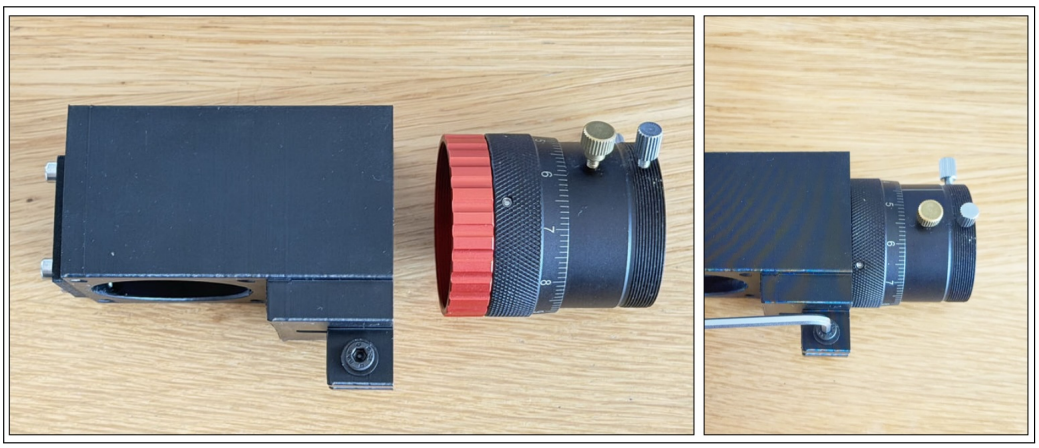


Figure 23. Fixing the camera interface, here a ZWO helical focuser.

A suggestion: fit the guiding camera with a filter that blocks the infrared (an 'IR-cut' filter, Figure 24). The detector is sensitive to infrared, but the lenses used perform only moderately well there, in particular because of chromatic aberration. If you observe from a town, this is also an opportunity to use a light-pollution filter: by increasing the contrast against the bright sky background, it will make faint objects easier to point at.

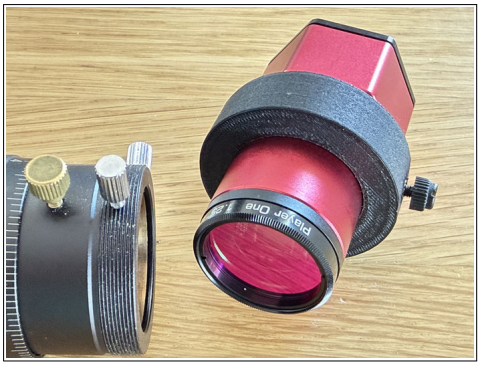


Figure 24. Guiding camera fitted with a UV and IR blocking filter ('UV/IR-cut').

Parts `1G-1-cube-guidage`, `2G-1-interface-guidage` and `8G-1-interface-telescope` are assembled with four M4 through-screws 70 mm long, as shown in Figure 25.

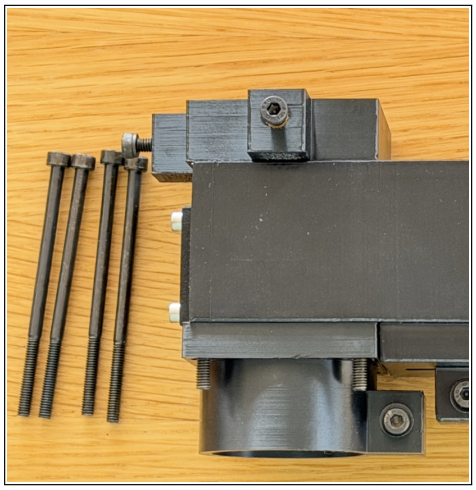


Figure 25. Assembling the guiding interface and the telescope interface on the guiding cube.

For greater rigidity, it is advisable to attach the spectrograph to the telescope with a 50 mm barrel. You can use the printed barrel, part `9G-1-coulant-50mm`, or a metal barrel combined with a male/female T2 ring 45 mm in diameter and 15 mm long (Figure 26).

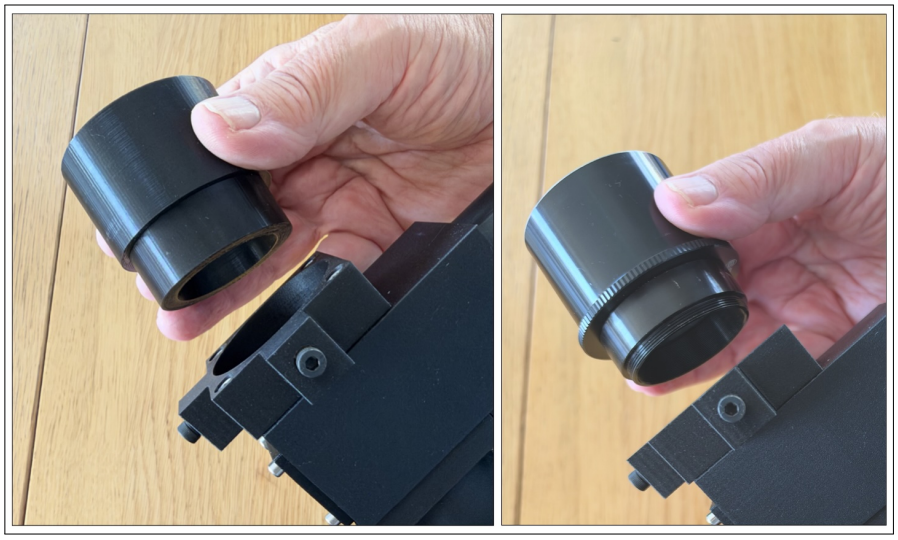


Figure 26. Fitting a 50 mm barrel to attach the spectrograph to the telescope. On the left, part `9G-1-coulant-50mm` provides this connection: a simple, effective and inexpensive solution.

Finally, the guiding system is coupled to the spectrograph through the 40 mm diameter interface (Figure 27).



Figure 27. The spectrograph about to receive its pointing and guiding system.

Push the guiding system fully home in the direction of the arrow shown in Figure 28. Then rotate it about the 40 mm interface, which allows this degree of freedom, so that the guiding housing lies in the same plane as the spectrograph; judging by eye is enough. Then firmly tighten the M4 × 20 mm screw of the spectrograph/guiding system interface.

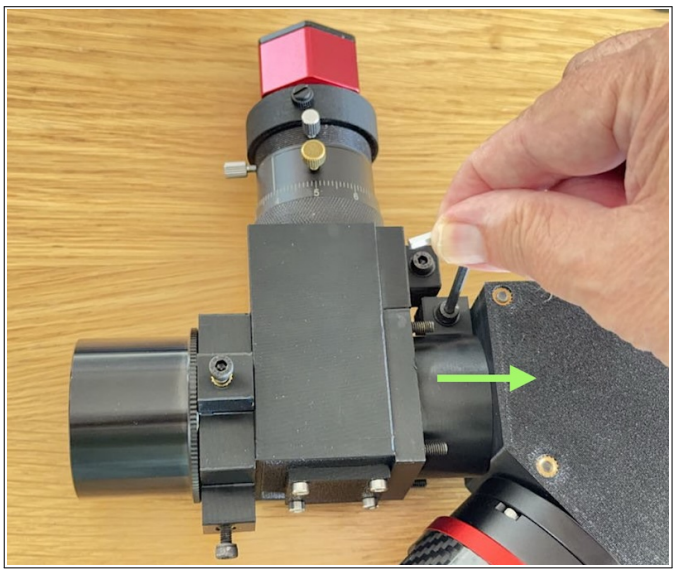


Figure 28. The TinyStar'Ex spectrograph fitted with its pointing and guiding system. Firmly tighten the M4 × 20 mm screw with an Allen key.

7 Adjustment

The only adjustment required is to orient the image of the slit correctly relative to the rows (or columns) of the guiding camera, and then to focus on this image.

A first piece of advice: if you use a telescope of less than 200 mm in diameter, work with the 14 µm slit of the GEN2 system (see the spectrograph assembly guide). From 200 mm up to 300 mm, select the 20 µm slit. For larger apertures, use the 26 µm slit. There is no absolute rule; these are only guidelines: the actual choice depends on the turbulence (*seeing*) at the observing site, the optical quality of your telescope, the accuracy of the tracking and the type of observing programme (depending on whether you favour spectral resolution or radiometric resolution, that is, signal-to-noise ratio). A refractor may be suitable, but a reflecting telescope, even of modest diameter, is always recommended in spectroscopy, as it is free of chromatic aberration.

A second piece of advice: for optimum adjustment when working on the bench (that is, without a telescope), feed the spectrograph with a beam whose geometry is roughly the same as it will have once the spectrograph is coupled to the telescope. What matters here is the opening angle of this beam, or focal ratio F/D. Optical aberrations will then be much better taken into account, and you will get a more accurate diagnosis of the state of your instrument.

Part `12G-1-simulateur-f4` is a simple diaphragm that fits on the front of the printed barrel `9G-1-coulant-50mm` and simulates the use of an f/4 telescope (Figure 29). The aperture is set by changing the diameter d of the diaphragm. If F/D is the focal ratio you wish to simulate, print a part whose diaphragm has a diameter, in millimetres, of d = 86/(F/D), where 86 mm is the distance between the plane of the diaphragm and the slit.

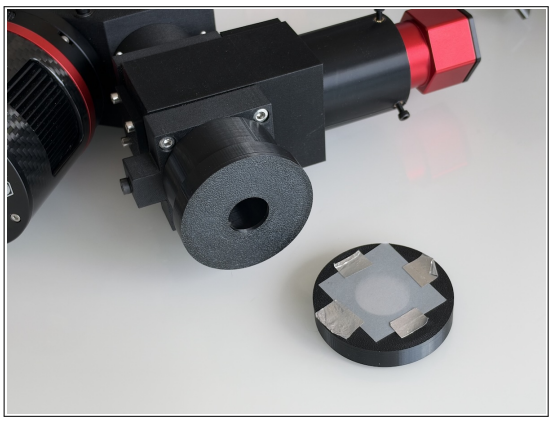


Figure 29. Fitting the entrance diaphragm to simulate a telescope of a given focal ratio. For an even better result, fix a sheet of tracing paper over the aperture to diffuse the light better. This accessory is useful for all adjustments, including those of the spectrograph itself (see the spectrograph assembly guide).

The aim is an image of the slit similar to that in Figure 30: the image is sharp, at least in its central part, the slit is horizontal and the image is centred vertically.

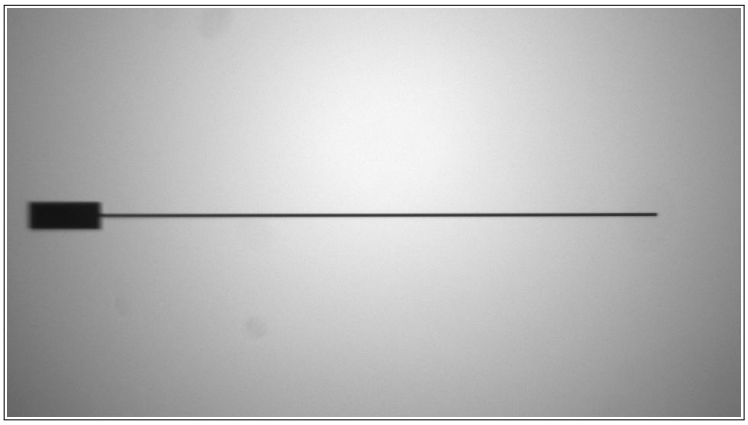


Figure 30. Image of the slit as seen by the guiding system.

For this adjustment, point the spectrograph at a lit ceiling, fitted if possible with its entrance diaphragm. If you intend to use a filter on the camera, fit it now. Display the camera image in 'live view' mode and adjust the exposure time so as not to saturate the image.

The illustrations in Figure 31 show how to proceed. Push the interface tube fully home in its seat, find the best focus by moving the camera forwards or backwards, then tighten the camera retaining screws. Next, orient the interface tube, and finally tighten the clamp of the guiding block.

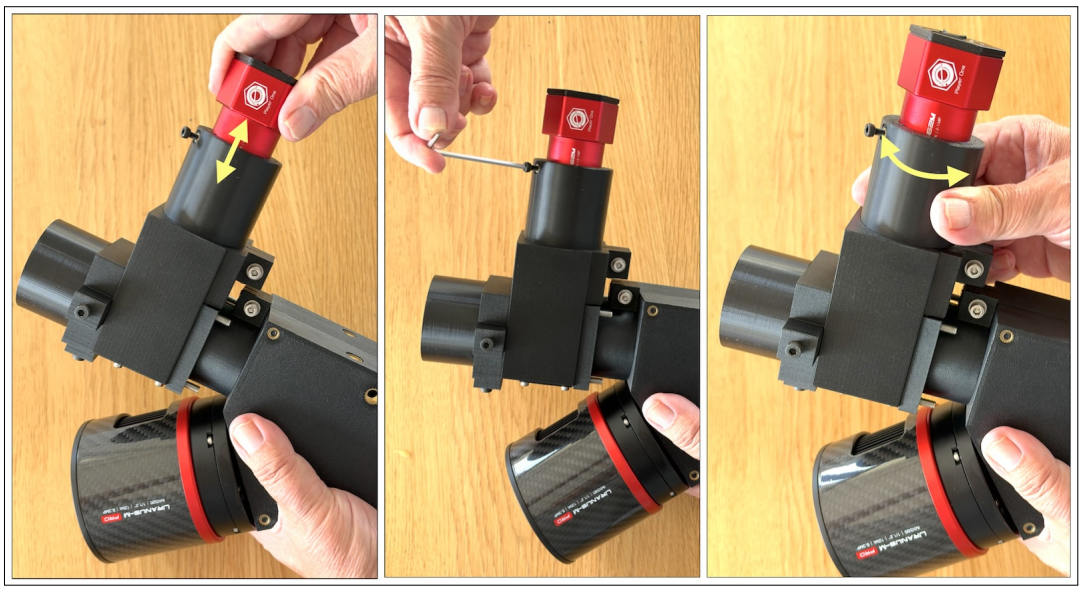


Figure 31. The operations required to adjust the image of the slit.

Work with a light touch, adjusting the tightness of the screws so that the parts can move without excessive play. After a few attempts and a little patience, you will achieve the precision of a helical focuser, with excellent stability into the bargain.

This adjustment should not normally need to be repeated. Bear in mind that your performance at the telescope will depend greatly on it. Enlarge the image of the slit on the screen and aim for really sharp edges. A slit oriented to within two or three tenths of a degree of the detector rows also makes observing much more comfortable. So take the time needed: a great deal depends on it.

If the image of the slit does not seem well enough centred vertically (a small offset is of no consequence), remove the mirror holder (`3G-1-support-miroir`) and add the shim `7G-1-entretoise-miroir` before refitting it. You will then need to refocus on the slit (see figure 32).

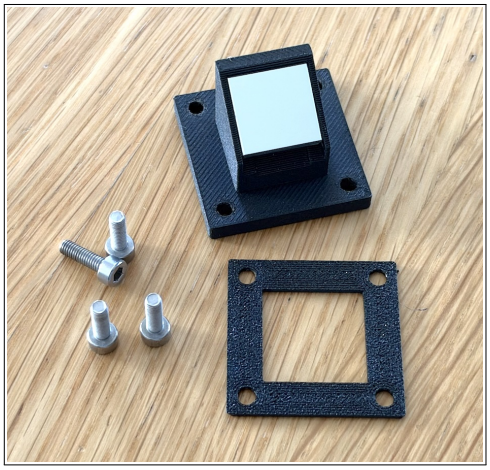


Figure 32. Shim for adjusting the position of the slit image.

TinyStar'Ex is now assembled and adjusted. Clear skies!

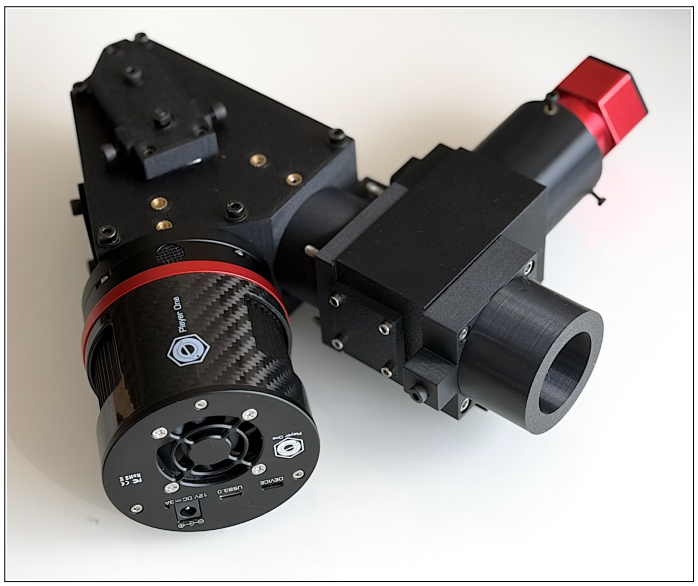


Figure 33. The complete `TinyStar'Ex` spectrograph, ready for use on the sky.